1) Given two integers N and M as inputs, print a rectangle of N * M stars.
For example, if N = 3, M = 4 then pattern will be like:

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****
****
****
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2) Take an integer N as input and print the count of its factors.
The factor of a number is the number that divides it perfectly leaving no remainder.
Example: 1, 2, 3, and 6 are factors of 6

3) Take an integer A as input, you have to tell whether it is a prime number or not.
A prime number is a natural number greater than 1 which is divisible only by 1 and itself.

4) Given an integer N as input, you have to tell whether it is a perfect number or not. A perfect number is a positive integer that is equal to the sum of its proper positive divisors (excluding the number itself).

A positive proper divisor divides a number without leaving any remainder.

5) Implement a program that takes two positive integers A and B in the input and prints their LCM.

Definition of LCM : LCM is the smallest or least positive integer that is divisible by both A and B.

6) You should take integer N as input. Your task is to calculate and print the sum of the digits of the number N.

7) You are given an integer N, Check the given number is a Armstrong Number or not.
For example, 153 = (1 * 1 * 1) + (5 * 5 * 5) + (3 * 3 * 3).

8) Given a number A. Return 1 if it is a perfect square otherwise, return 0.

Example : 4 is a perfect square, 1001 is not a perfect square.

9) Given a sorted integer array A, and an integer B. Find the first and last index of B in A.

A = [-2, -2, 4, 4, 8, 9]

B = 4

Output : C = [2, 3]

A = [1, 9, 9, 9, 10, 21]

B = 9

Output : C = [1, 3]

10) You are given an integer array A and an integer B.

You are required to return the count of pairs having sum equal to B.
Note : i!=j(Pair Number indexes should not be same)

A = [1,2,3,2,1]   B = 5

Output : 2

A = [1,1,1]   B = 2

Output : 3

11) You are given an integer array A. Now your task is to find the inverse of A.

Now, the inverse of the array is A will be an array in which we change the

positions of the values as their indices and indices as values.

So, array A = [2, 0, 1]

- Now 2 is at index 0. So, place 0 at index 2.
- 0 is at index 1. So, place 1 at index 0.
- 1 is at index 2. So, place 2 at index 1.

So, the inverse of A will be [1, 2, 0]

12) Given a 2D integer array C[][] of A rows and B columns. Return an integer array of size B that represents the sums of the columns of the 2D array C.

A = 3

B = 2

C = [[4, 1], [1, 3], [6, 2]]

Output : [11, 6]

13) Write a program to input an integer N and a N*N matrix Mat from user and print the matrix in wave form (column wise)

See example for clarifications regarding wave print.

Note: Ensure there is a space character (' ') at the end of the line.

3

4 1 2

7 4 4

3 7 4

Output :

4 7 3 7 4 1 2 4 4

14) Given an Array of integers A, every element in it is repeated twice except one, find that unique element.

A = [1, 4, 3, 5, 2, 3, 5, 1, 4]

Output : 2

15) Write a program that returns the list of elements that are present in the given list and are divisible by 5 and 7.

Note : Use ArrayList.

Input : A = [5, 7, 70, 50, 35]

Output : [70, 35]

16) Write a Java program to demonstrate method overloading and method overriding with an example.

17) Print numbers 1 to 10 in a separate Thread.

18) Write a program to find if a string is a palindrome using OOP principles in Java.

19) **Create a Banking Application**

Problem Statement:

Simulate basic banking operations.

Key Requirements:

Define a class Account with methods for deposit and withdrawal.

Create subclasses like SavingsAccount and CurrentAccount with specific rules (e.g., interest rates, overdraft limits).

Implement a method to display account details.

20) **Build an Employee Management System**

Problem Statement:

Develop a system to manage employees in a company.

Key Requirements:

Create a base class Employee with common attributes like name and ID.

Derive classes like Manager, Developer, and Intern from Employee.

Implement methods to calculate salary, which varies based on the employee type.

21) **Implement a Functional Interface for Arithmetic Operations**

Define a functional interface Operation with a method int execute(int a, int b).

Implement addition, subtraction, multiplication, and division using lambda expressions.

22) Create a Spring Boot application that uses @RequestMapping to handle HTTP GET, POST, PUT, and DELETE requests.

23) Create an interface called Vehicle with methods like start(), stop(), and accelerate(). Then, create classes Car and Bike that implement the Vehicle interface.

24) Create a React component that displays a welcome message using props.

25) Build a counter app using useState to increment and decrement a number.

26) How do you handle the onChange event for an input box in React?

27) How do you render a list of fruits from an array using map()?

28) Implement a Toggle Switch

Task: Develop a ToggleSwitch component that changes its state between "On" and "Off" when clicked.

29) Create a form with an input field and a submit button. Display the entered text below the form upon submission.

30) Implement a simple calculator using React, that adds two numbers entered by the user.